

Field Ontologies of the RSVP Stack: From Scalar Smoothing to Cognitive Closure (Revised)

Flyxion

November 12, 2025

The universe is not made of things, but of processes.
Alfred North Whitehead [14]

Abstract

We present a unified field-theoretic framework (the RSVP stack) that coherently models cosmology, semantics, institutional closure, discrete memory lattices, and cognitive optimization through a single scalarvectorentropy triplet $(\Phi, \mathbf{v}, S)$. We give a formal action functional from which the RSVP dynamics are derived, define coarse-grainings that yield EBSSC, Spherpap, TARTAN, and CLIO, and provide stability criteria, conserved quantities, and failure-mode taxonomy. The manuscript includes schematic TikZ diagrams for the stack architecture, field flow, sphere merges, and lattice refinement. We conclude by proposing computational experiments and observational tests to validate entropic redshift and agency-coherence predictions.

1 The Problem of Fragmented Ontologies

A recurring limitation across physics, cognition, and social theory is the disciplinary partitioning of formalisms: cosmology treats metric expansion, cognitive science treats symbolic computation, and social theory treats institutions as distinct ontologies. This partitioning obstructs the identification of shared dynamical principles that govern structure formation and persistence at multiple scales. The present work addresses this limitation by proposing a unified field-theoretic substrate in which scalar, vector, and entropic components interact to generate emergent phenomena across these domains.

We introduce the *Relativistic ScalarVector Plenum* (RSVP) as the foundational layer, comprising a scalar field Φ that encodes potential meaning or density, a vector field $\mathbf{v}$ that represents directed flow or agency, and an entropy field S that quantifies local disorder and imposes thermodynamic constraints [13, 4]. The derived *RSVP stack* consists of five hierarchically coupled levels: the base RSVP dynamics, the Entropy-Bounded Sparse Semantic Calculus (EBSSC), the Spherpap Calculus for institutional closure, the TARTAN recursive lattices for memory representation, and the CLIO cognitive descent for optimization. These levels are not independent theories but rather distinct coarse-grainings and reinterpretations of the same underlying field equations, achieved through observational functors that preserve key invariants [1].

The objectives of this manuscript are threefold. First, we derive the RSVP field equations from a universal variational principle expressed as an action functional. Second, we demonstrate how semantic manifolds, institutional boundaries, mnemonic structures, and cognitive policies

emerge as structured projections of these dynamics. Third, we establish quantitative stability criteria, conserved quantities, and a taxonomy of failure modes that apply uniformly across layers. To ensure clarity, all primary relations are presented in boxed equations with explicit labels for cross-referencing. Schematic diagrams rendered in TikZ illustrate the hierarchical architecture, local interactions, and transformation processes.

This framework yields testable predictions, including an entropic mechanism for apparent cosmic redshift and coherence metrics for agency persistence. It also suggests novel computational architectures that incorporate entropy budgets and sparsity constraints directly into learning objectives, echoing active inference principles [5].

Theoretical Context

The RSVP stack draws from multiple lineages: non-equilibrium thermodynamics [13, 7], information geometry and active inference [6], higher category theory [9], and agential realism [3]. It extends field-theoretic unification attempts [2] into semantic and social domains, treating meaning not as epiphenomenal but as a conserved density subject to thermodynamic law [4].

2 Mathematical Vocabulary of the Stack

Let $\Omega \subset \mathbb{R}^d$ ($d = 3$ typically) be an open, bounded domain with Lipschitz boundary $\partial\Omega$. We work in Sobolev spaces: $H^k(\Omega) = W^{k,2}(\Omega)$, with dual $H^{-k}(\Omega)$. The fields inhabit:

$$\Phi \in H^1(\Omega), \quad \mathbf{v} \in [H^1(\Omega)]^d, \quad S \in L^2(\Omega).$$

Initial data satisfy:

$$\Phi_0 \in H^1(\Omega), \quad \mathbf{v}_0 \in [H^1(\Omega)]^d, \quad S_0 \in L^2(\Omega) \cap L^\infty(\Omega), \quad S_0 \geq 0.$$

Boundary conditions are homogeneous Neumann:

$$\partial_n \Phi = 0, \quad \partial_n \mathbf{v} = 0, \quad \partial_n S = 0 \quad \text{on } \partial\Omega.$$

Φ Scalar potential / semantic density.

$\mathbf{v}$ Vector flow field (agency).

S Entropy field (local disorder / expressivity budget).

$\mathcal{L}$ Loss functional (CLIO).

$\mathcal{A}$ Action functional (formal structure).

Γ Agency coherence metric.

3 The Grammar of Dynamics

At the heart of the plenum lies a principle as old as mechanics itself: equilibrium through variation. The RSVP action functional extends this heritage by balancing energy, meaning, and entropy in a single integral form, unifying Hamiltonian mechanics with thermodynamic necessity [11].

The dynamics are derived from the augmented action:

$$\mathcal{A}[\Phi, \mathbf{v}, S] + \mathcal{D}[\Phi, \mathbf{v}] = \int_{t_0}^{t_1} \int_{\Omega} \left(\frac{1}{2} \|\mathbf{v}\|^2 + U(\Phi) - T(S) + \mathcal{R}(\Phi, \mathbf{v}, S) \right) dx dt + \frac{1}{2} \int_{t_0}^{t_1} \int_{\Omega} (\kappa_{\Phi} |\nabla \Phi|^2 + \kappa_v |\nabla \mathbf{v}|^2) dx dt, \quad (1)$$

where $\mathcal{D}$ is the Rayleigh dissipation functional.

The action does not describe particles in space, but processes of becoming where meaning flows, resists, and remembers [3].

§1 Function Spaces, Regularity, and Well-Posedness of the RSVP Fields

Assumption 3.1. The potential $U \in C^2(\mathbb{R})$ with $U'' \geq 0$, $\sigma(\Phi, \mathbf{v}) \leq C(1 + |\Phi| + \|\mathbf{v}\|^2)$, and $\kappa_{\Phi}, \kappa_v, \kappa_S > 0$.

Theorem 3.2 (Local Classical Solutions). *Under theorem 3.1, there exists $T^* > 0$ depending on $\|\Phi_0\|_{H^1}$, $\|\mathbf{v}_0\|_{H^1}$, $\|S_0\|_{L^\infty}$ such that the RSVP system admits a unique classical solution*

$$\Phi \in C([0, T^*]; H^2(\Omega)) \cap C^1([0, T^*]; H^1(\Omega)),$$

and similarly for $\mathbf{v}$, S .

Proof Sketch. Rewrite the system as

$$\partial_t u = \mathcal{L}u + \mathcal{N}(u), \quad u = (\Phi, \mathbf{v}, S),$$

where $\mathcal{L}$ is the sectorial operator generated by $(-\kappa_{\Phi}\Delta, -\kappa_v\Delta, -\kappa_S\Delta + \lambda)$ with Neumann conditions. The nonlinearity $\mathcal{N}$ is locally Lipschitz from $H^2 \times [H^2]^d \times H^1 \rightarrow H^1 \times [H^1]^d \times L^2$ by Sobolev embedding and Assumption 3.1. Apply the analytic semigroup theorem [12, Thm. 3.3.3]. $\square$

Theorem 3.3 (Global Weak Solutions). *There exists a global weak solution in the LerayHopf sense:*

$$\Phi \in L^\infty(0, \infty; H^1) \cap L_{loc}^2(0, \infty; H^2), \quad \partial_t \Phi \in L_{loc}^2(0, \infty; H^{-1}),$$

and analogously for $\mathbf{v}$, S .

Proof Sketch. Galerkin approximation with eigenfunctions of $-\Delta$ (Neumann). Energy estimates:

$$\frac{d}{dt} \left(\frac{1}{2} \int \Phi^2 + \frac{1}{2} \int \|\mathbf{v}\|^2 + \int S \right) \leq C,$$

using $\sigma \leq C(1 + |\Phi| + \|\mathbf{v}\|^2)$ and Youngs inequality. AubinLions compactness yields strong convergence in $L^2(0, T; L^2)$ for passage to the limit [8]. $\square$

Corollary 3.4 (Falsifiable Breakdown). *If numerical simulations exhibit finite-time blow-up in $\|\nabla \Phi\|_{L^\infty}$ despite $\kappa_{\Phi} > 0$, then higher-order regularization (e.g., $\epsilon \Delta^2 \Phi$) is required.*

§2 Variational Derivation and Dissipation

Theorem 3.5 (Euler–Lagrange with Dissipation). *Stationary points of (1) yield the RSVP field equations (2)–(4).*

Proof Sketch. Variation in Φ :

$$\delta_\Phi(\mathcal{A} + \mathcal{D}) = \int (U'(\Phi) - \nabla \cdot (\Phi \mathbf{v}) - \kappa_\Phi \Delta \Phi) \delta \Phi = 0.$$

Variation in $\mathbf{v}$:

$$\delta_{\mathbf{v}}(\mathcal{A} + \mathcal{D}) = \int (\mathbf{v} + \Phi \nabla \Phi - (\mathbf{v} \cdot \nabla) \mathbf{v} - \kappa_v \Delta \mathbf{v}) \cdot \delta \mathbf{v} = 0.$$

The entropy equation arises from a separate Onsager balance with production σ . □

§3 Entropy Bound and Lyapunov Structure

Lemma 3.6 (Global Entropy Bound). *Under theorem 3.1,*

$$\int_\Omega S(x, t) dx \leq B_S := \max \left\{ \int_\Omega S_0, \frac{\text{ess sup } \sigma}{\lambda} |\Omega| \right\}.$$

Proof. Integrate (4):

$$\frac{d}{dt} \int S = \int \sigma - \lambda \int S + \kappa_S \int \Delta S = \int \sigma - \lambda \int S.$$

Apply Grönwall: $\int S(t) \leq e^{-\lambda t} \int S_0 + \frac{\text{ess sup } \sigma}{\lambda} (1 - e^{-\lambda t})$. □

Theorem 3.7 (Lyapunov Functional). *Define $\mathcal{H}(t) = \int_\Omega S dx$. Then $\dot{\mathcal{H}} \leq \text{ess sup } \sigma - \lambda \mathcal{H}$.*

§4 Asymptotic Smoothing (Lamphrodyne)

Theorem 3.8 (Exponential Gradient Decay). *If $\sigma \equiv 0$ on $[T, \infty)$, then*

$$\|\nabla S(t)\|_{L^2} \leq \|\nabla S(T)\|_{L^2} e^{-\mu(t-T)}, \quad \mu = \min(\lambda, \kappa_S \lambda_1),$$

where $\lambda_1 > 0$ is the first Neumann eigenvalue of $-\Delta$.

Proof. Multiply (4) by $-\Delta S$, integrate:

$$\frac{1}{2} \frac{d}{dt} \int |\nabla S|^2 + \kappa_S \int |\Delta S|^2 + \lambda \int |\nabla S|^2 = 0.$$

Poincaré: $\int |\Delta S|^2 \geq \lambda_1 \int |\nabla S|^2$. □

4 The Laws Beneath the Stack

From the variational grammar emerges a living syntax: three coupled equations that govern how meaning moves, persists, and pays its entropic toll.

$$\boxed{\partial_t \Phi = -\nabla \cdot (\Phi \mathbf{v}) + \kappa_\Phi \Delta \Phi.} \tag{2}$$

$$\boxed{\partial_t \mathbf{v} = -(\mathbf{v} \cdot \nabla) \mathbf{v} - \nabla \Phi + \kappa_v \Delta \mathbf{v},} \tag{3}$$

$$\boxed{\partial_t S = \sigma(\Phi, \mathbf{v}) + \kappa_S \Delta S - \lambda S.} \tag{4}$$

§5 Conservation Laws

Theorem 4.1 (Scalar Conservation).

$$\frac{d}{dt} \int_{\Omega} \Phi \, dx = 0.$$

Proof. Integrate (2) and apply divergence theorem with Neumann conditions. $\square$

Quantity	Expression
Scalar mass	$\int \Phi$ (conserved)
Kinetic energy	$\frac{1}{2} \int \ \mathbf{v}\ ^2$
Entropy budget	$\int S \leq B_S$
Helicity (if $d = 3$)	$\int \mathbf{v} \cdot (\nabla \times \mathbf{v})$

Table 1: Conserved and bounded quantities.

5 Measuring Meaning and Coherence

$$\mathcal{C}(t) = \int_{\Omega} \Phi(x, t) \|\nabla \times \mathbf{v}(x, t)\| \, dx. \quad (5)$$

$$z_{\text{RSVP}}(t_0, t_1) = \exp \left(\alpha_S \int_{t_0}^{t_1} \frac{1}{|\Omega|} \int_{\Omega} \partial_t S \, dx \, dt \right) - 1. \quad (6)$$

§6 Evolution of Coherence

Lemma 5.1.

$$\frac{d\mathcal{C}}{dt} \leq \int \Phi (|\nabla \times (-\nabla \Phi)| + \kappa_v |\nabla \times \Delta \mathbf{v}|) + \text{transport terms}.$$

6 The Semantics of Economy

When we zoom into the cognitive scale, the fields become semantic manifolds. Inference is no longer free; it must be *economical* sparse, bounded, and flow-preserving [6].

The semantic manifold is the triplet:

$$\mathcal{M} = (\Phi, \mathbf{v}, S). \quad (7)$$

A sparsity prior enforces economical representation:

$$\|\Phi\|_0 \ll N, \quad (8)$$

where $N = |\Omega|$ in discretized form.

Optimal policies minimize a composite objective:

$$\pi^* = \arg \min_{\pi} \mathbb{E} \left[S + \alpha \|\Phi\|_0 + \beta \|\nabla \cdot \mathbf{v}\|^2 \right]. \quad (9)$$

§7 EBSSC as Entropy-Constrained Sparse Optimization

Assumption 6.1. The action space $\mathcal{A}$ is finite, $\Phi \in \ell^0(\mathcal{A})$, and $\nabla \cdot \mathbf{v}$ is approximated via finite differences. The entropy S is the Shannon entropy of the policy π .

Theorem 6.2 (Existence of Optimal Sparse Policy). *Under theorem 6.1, there exists a minimizer $\pi^* \in \mathcal{P}(\mathcal{A})$ satisfying (9).*

Proof Sketch. The functional $J(\pi) = \mathbb{E}_\pi[S] + \alpha \|\Phi\|_0 + \beta \mathbb{E}_\pi[\|\nabla \cdot \mathbf{v}\|^2]$ is lower semicontinuous on the compact simplex $\mathcal{P}(\mathcal{A})$ (weak-* topology). The ℓ^0 term is relaxed via L^1 proxy in practice, but direct minimization over sparse supports yields existence by finite dimensionality. $\square$

Theorem 6.3 (Sparsity-Entropy Tradeoff Bound).

$$\|\Phi\|_0 \leq \frac{\mathbb{E}[S] + \beta \mathbb{E}[\|\nabla \cdot \mathbf{v}\|^2]}{\alpha}.$$

Proof. From (9), at optimum $J(\pi^*) \geq \alpha \|\Phi\|_0$. $\square$

Corollary 6.4 (Falsifiable Prediction). *If empirical cognitive policies (e.g., human decision traces) violate the bound with α calibrated from neural firing rates, then additional latent variables must be included.*

7 Thought as Sparse Flow

Cognition is not a static map but a *policy*: a sparse distribution over possible acts, updated in light of evidence.

Cognitive policies are sparse distributions over actions:

$$\boxed{\pi(a|\Phi, \mathbf{v}, S), \quad |\text{supp}(\pi)| \ll |\mathcal{A}|.} \quad (10)$$

Bayesian updates proceed in the manifold:

$$\boxed{p(\Phi|D) \propto p(D|\Phi)p(\Phi).} \quad (11)$$

The agency coherence ratio is

$$\boxed{\Gamma(t) = \frac{I(\Phi; \mathbf{v})}{H(S)}.} \quad (12)$$

§8 Mutual Information Bounds on Agency Coherence

Assumption 7.1. The fields are jointly Gaussian in local patches, or sub-Gaussian with bounded variance.

Theorem 7.2 (Upper Bound on Coherence).

$$\Gamma(t) \leq \frac{1}{2} \log \left(1 + \frac{\text{Var}(\Phi)}{\text{Var}(S)} \cdot \frac{\text{Cov}(\Phi, \mathbf{v})^2}{\text{Var}(\mathbf{v})} \right).$$

Proof Sketch. By data processing inequality and Gaussian mutual information formula:

$$I(\Phi; \mathbf{v}) \leq \frac{1}{2} \log \det(\Sigma_{\Phi\mathbf{v}}) - \frac{1}{2} \log \det(\Sigma_\Phi) \det(\Sigma_{\mathbf{v}}).$$

Under theorem 7.1, this reduces to the correlation form. $\square$

Theorem 7.3 (Lower Bound via Flow Alignment). *If $\mathbf{v} = -\nabla \log p(\Phi|D)$, then*

$$\Gamma(t) \geq \frac{\mathbb{E}[\|\nabla \log p\|^2]}{H(S)}.$$

Proof. Direct from definition and Fisher information. $\square$

Corollary 7.4 (Neuroimaging Falsifier). *If fMRI BOLD signal correlation between semantic areas and motor planning exceeds Γ predicted from entropy estimates (via EEG microstate entropy), the model requires nonlocal interactions.*

8 Institutions as Thermodynamic Spheres

Societies are bounded thermodynamic systems maintaining interior order via ritual and cryptographic hardness [4].

Institutions are modeled as spheres:

$$\boxed{\mathcal{S} = (I, B, \Sigma).} \quad (13)$$

The pop operation merges spheres:

$$\boxed{P : \mathcal{S}_i \otimes \mathcal{S}_j \rightarrow \mathcal{S}_k.} \quad (14)$$

Merge occurs above threshold:

$$\boxed{\Sigma_i + \Sigma_j > \Theta_{\text{closure}}.} \quad (15)$$

Resistance:

$$\boxed{R = R_t + R_c = \int_0^T \gamma(t) dt + 2^H.} \quad (16)$$

§9 Spherepop Merge Stability and Resistance Threshold

Assumption 8.1. Permeability $\Sigma \in [0, 1]$, ritual cost rate $\gamma(t) \geq \gamma_{\min} > 0$, keyspace entropy $H \geq 0$.

Theorem 8.2 (Merge Condition). *Merge succeeds if and only if*

$$\Sigma_i + \Sigma_j > \Theta_{\text{closure}} \quad \text{and} \quad R_t + 2^H < P_{\max}.$$

Proof. By definition of P and anti-admissibility. $\square$

Theorem 8.3 (Non-Flattening Merge Preserves Volume). *For $\mathcal{S}_a \oplus \mathcal{S}_b$,*

$$|I_a \times I_b| = |I_a| \cdot |I_b|, \quad \Sigma_{\oplus} = \min(\Sigma_a, \Sigma_b).$$

Proof. Cartesian product and boundary intersection. $\square$

Lemma 8.4 (Resistance Growth).

$$\frac{dR}{dt} \geq \gamma_{\min} + \log_2(1 + \delta H(t)),$$

for incremental keyspace hardening.

Corollary 8.5 (Historical Falsifier). *If pre-modern institutions (low H) merged faster than predicted by ritual cost $\int \gamma(t)dt$, then social contagion terms must be added to Σ .*

9 Memory Lattices and Recursive Continuity

Memory is a *recursive weave*: each tile contains a microcosm of the field, refined across scales.

Memory tiles:

$$T = (\Phi_T, \mathbf{v}_T, S_T, m_T). \quad (17)$$

Refinement:

$$T_{n+1} = f(T_n), \quad f = \text{refine} + \text{annotate} + \text{smooth}. \quad (18)$$

Semantic noise:

$$\eta_s \sim \mathcal{N}(0, \sigma_S(\nabla\Phi)). \quad (19)$$

§10 TARTAN Refinement Convergence

Assumption 9.1. The refinement operator f is contractive in L^2 on each tile: $\|f(T) - T^*\|_{L^2} \leq \rho\|T - T^*\|_{L^2}$, $\rho < 1$.

Theorem 9.2 (Fixed-Point Convergence). *Under theorem 9.1, the lattice sequence converges exponentially:*

$$\|T_n - T^*\|_{L^2} \leq \rho^n \|T_0 - T^*\|_{L^2}.$$

Proof. Standard contraction mapping theorem in complete metric space $(L^2(\Omega), \|\cdot\|_{L^2})$. $\square$

Theorem 9.3 (Gradient-Weighted Noise Optimality). *The variance $\sigma_S(\nabla\Phi) = c|\nabla\Phi|^2$ maximizes exploration near decision boundaries.*

Proof Sketch. By information geometry, Fisher information is concentrated where $|\nabla \log p| > 0$. Noise scaled to gradient explores the Riemannian volume of the posterior. $\square$

Corollary 9.4 (Machine Learning Falsifier). *If gradient-weighted noise in TARTAN-based transformers yields worse OOD generalization than uniform noise, then the manifold hypothesis is violated in the data domain.*

10 Descent into Cognitive Optimization

Intelligence is a *descent*: a disciplined fall toward lower disorder.

Optimization:

$$\Phi_{t+1} = \Phi_t - \eta_t \nabla_{\Phi} \mathcal{L}(\Phi_t, \mathbf{v}_t, S_t). \quad (20)$$

The loss:

$$\mathcal{L} = \mathbb{E}[S] + \lambda_{\text{sparse}} \|\Phi\|_0 - \lambda_{\text{flow}} \|\mathbf{v}\|_2. \quad (21)$$

§11 CLIO Descent Convergence and Saddle Avoidance

Assumption 10.1. The loss $\mathcal{L}$ is μ -strongly convex in Φ on sparse supports, and $\nabla_{\Phi} \mathcal{L}$ is L -Lipschitz.

Theorem 10.2 (Linear Convergence). *With adaptive $\eta_t = 1/L$,*

$$\mathcal{L}(\Phi_t) - \mathcal{L}^* \leq (1 - \mu/L)^t (\mathcal{L}(\Phi_0) - \mathcal{L}^*).$$

Proof. Standard descent lemma for smooth strongly convex functions [10]. $\square$

Theorem 10.3 (Flow Reward Prevents Collapse). *The term $-\lambda_{\text{flow}}\|\mathbf{v}\|_2$ ensures $\mathbf{v} \not\rightarrow 0$ if $\lambda_{\text{flow}} > 0$, preventing stagnation at local minima with zero agency.*

Proof. Suppose $\mathbf{v}_t \rightarrow 0$. Then $\partial_t \mathcal{L} \leq \mathbb{E}[S] + \lambda_{\text{sparse}}\|\Phi\|_0$, but from (3), $\mathbf{v}$ is driven by $-\nabla\Phi$, contradicting collapse unless Φ is flat. $\square$

Corollary 10.4 (Behavioral Falsifier). *If ablating the flow reward in CLIO-trained agents causes policy collapse (high entropy, low action variance), the agency term is necessary. Observable in RL benchmarks (e.g., MuJoCo).*

11 Universal Patterns of Collapse

Across all scales, the same specters appear: gradient death, flow stagnation, entropy saturation. These are *universal failure modes* of structured becoming.

Layer	Interpretation	Failure signature
RSVP	Physical potential/flow/entropy	$\nabla\Phi \rightarrow 0, \mathbf{v} \rightarrow 0, S \rightarrow B_S$
EBSSC	Semantic sparsity	$\ \Phi\ _0 \rightarrow N, H(S) \uparrow$
Spherepop	Institutional boundary	$\Sigma \uparrow, R \downarrow$ collapse
TARTAN	Memory lattice	tile decoherence, loss of annotations m_T
CLIO	Cognitive optimizer	gradient vanishing / oversmoothing

Table 2: Unified interpretation across scales and shared failure signatures.

§12 Unified Failure Modes and Entropy Saturation

Assumption 11.1. The system evolves under theorem 3.1 with $\sigma(\Phi, \mathbf{v}) \leq K(|\Phi| + \|\mathbf{v}\|^2)$ and $\lambda > 0$.

Theorem 11.2 (Entropy Saturation Bound).

$$\limsup_{t \rightarrow \infty} \int_{\Omega} S(x, t) dx \leq \frac{K}{\lambda} \left(\int_{\Omega} \Phi_0 + \sup_t \int_{\Omega} \frac{1}{2} \|\mathbf{v}\|^2 \right).$$

Proof Sketch. From ?? and energy estimates on kinetic term via integration of (3) multiplied by $\mathbf{v}$:

$$\frac{d}{dt} \int \frac{1}{2} \|\mathbf{v}\|^2 + \kappa_v \int |\nabla \mathbf{v}|^2 = - \int \mathbf{v} \cdot \nabla \Phi.$$

Bound via Cauchy–Schwarz and Poincaré, then substitute into $\int \sigma$. $\square$

Theorem 11.3 (Gradient Flattening). *If $\sigma \rightarrow 0$ pointwise and $\kappa_{\Phi}, \kappa_v > 0$, then*

$$\|\nabla\Phi(t)\|_{L^2} + \|\nabla\mathbf{v}(t)\|_{L^2} \rightarrow 0 \quad \text{exponentially.}$$

Proof. Energy method on $\int |\nabla\Phi|^2$ and $\int |\nabla\mathbf{v}|^2$, using vanishing source and diffusion dominance. $\square$

Corollary 11.4 (Cosmological Falsifier). *If galaxy cluster voids show persistent scalar gradients (via weak lensing) despite low σ , then nonlocal coupling or negative κ_{Φ} (anti-diffusion) is required.*

12 Empirical Horizons: Entropic Redshift

The entropic redshift z_{RSVP} offers a falsifiable alternative to metric expansion.

$$z_{\text{RSVP}}(t_0, t_1) = \exp \left(\alpha_S \int_{t_0}^{t_1} \frac{1}{|\Omega|} \int_{\Omega} \partial_t S \, dx \, dt \right) - 1. \quad (22)$$

§13 Entropic Redshift as Cumulative Entropy Flux

Assumption 12.1. Entropy production $\sigma = \sigma_0(\nabla \Phi \cdot \mathbf{v})^2$ is quadratic in shear, and $\alpha_S > 0$ is a universal calibration.

Theorem 12.2 (Redshift Monotonicity). *Under theorem 12.1, $z_{\text{RSVP}}(t_0, t)$ is non-decreasing in t .*

Proof. Since $\partial_t S \geq -\lambda S + \kappa_S \Delta S \geq -\lambda B_S$, but dominant term is $\int \sigma \geq 0$, so $\int \partial_t S > 0$ on average. $\square$

Theorem 12.3 (Scaling Relation). *For self-similar evolution with $\Phi(t) = \phi(t/R(t))$, $\mathbf{v}(t) = v(t/R(t))$, then*

$$z_{\text{RSVP}} \sim \alpha_S \sigma_0 \frac{\int \phi v \, dt}{R(t)^2}.$$

Proof Sketch. Change of variables in double integral, using similarity ansatz. $\square$

Corollary 12.4 (Observational Falsifier). *Fit supernova Ia light curves to $z_{\text{RSVP}}(t)$ vs. $z_{\Lambda\text{CDM}}$. If $\chi_{\text{RSVP}}^2 > \chi_{\Lambda\text{CDM}}^2 + 5\sqrt{2N}$ (Wilks), reject entropic mechanism at 2σ .*

13 Categorical Coarse-Graining of the Stack

The RSVP stack is not a metaphor but a *functorial hierarchy*: each layer is a category, each transition a structure-preserving map.

§14 Functorial Invariants and Coarse-Graining

Assumption 13.1. Each layer $\mathcal{L}_i$ is a category with objects $(\Phi, \mathbf{v}, S)$ on different lattices, morphisms field transformations.

Theorem 13.2 (Entropy Monotonicity Preserved). *Each functor $F_i : \mathcal{L}_i \rightarrow \mathcal{L}_{i+1}$ satisfies*

$$H(F_i(S)) \leq H(S) + C_i,$$

with C_i a bounded coarsening constant.

Proof Sketch. Coarse-graining via conditional expectation: $F_i(S) = \mathbb{E}[S|\mathcal{G}_{i+1}]$, where $\mathcal{G}_{i+1}$ is coarser σ -algebra. By Jensen and convexity of entropy. $\square$

Theorem 13.3 (Sparsity Invariant). $\|\Phi\|_0$ is non-increasing under $F_1 : \text{RSVP} \rightarrow \text{EBSSC}$.

Proof. EBSSC imposes hard sparsity prior; coarse-graining projects onto sparse basis. $\square$

Corollary 13.4 (Theoretical Falsifier). *If a cognitive model requires denser representations at higher abstraction (e.g., concepts > pixels), then F_1 must be replaced by a lax functor allowing sparsity growth.*

14 Linear Stability and Bifurcation

Linearize around homogeneous equilibrium $(\Phi_*, \mathbf{v}_* = 0, S_*)$.

§15 Linear Stability of Homogeneous State

Assumption 14.1. Equilibrium: $\nabla\Phi_* = 0$, $\mathbf{v}_* = 0$, $\sigma(\Phi_*, \mathbf{v}_*) = 0$, $S_* = B_S/\lambda$.

Theorem 14.2 (Dispersion Relation). *For perturbations $\sim e^{\lambda t + i\mathbf{k}\cdot\mathbf{x}}$, the growth rate satisfies*

$$\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0 = 0,$$

with

$$a_2 = (\kappa_\Phi + \kappa_v + \kappa_S)k^2 + \lambda, \quad a_0 = \kappa_\Phi\kappa_v\kappa_S k^6.$$

All roots have $\text{Re}(\lambda) < 0$ for $k \neq 0$.

Proof Sketch. Fourier transform the linearized system:

$$\partial_t \hat{\Phi} = -ik\Phi_* \hat{v}_x - \kappa_\Phi k^2 \hat{\Phi},$$

etc. Characteristic polynomial follows from 3×3 Jacobian. $\square$

Theorem 14.3 (Turing Instability Threshold). *If $\kappa_v \ll \kappa_\Phi$, long-wavelength instability emerges when*

$$\Phi_* > \frac{\kappa_v(\kappa_\Phi + \kappa_S)}{\kappa_\Phi}.$$

Proof. Routh–Hurwitz failure at $k \rightarrow 0$. $\square$

Corollary 14.4 (Cosmological Structure Formation Falsifier). *If void sizes grow faster than predicted by Φ_* -driven instability, then active matter terms (self-propulsion) must be added to $\mathbf{v}$.*

15 Numerical Schemes and Computational Validation

The RSVP system is simulated via operator splitting and finite volumes.

§16 Numerical Consistency, Stability, and CFL Condition

Assumption 15.1. Spatial discretization: uniform grid $\Delta x = h$, time step Δt . Second-order finite differences, implicit diffusion, explicit advection.

Theorem 15.2 (CFL Condition). *For stability of explicit advection,*

$$\Delta t \leq \frac{h}{\max(|\mathbf{v}_x|, |\mathbf{v}_y|, |\mathbf{v}_z|)} =: \Delta t_{\text{adv}}.$$

For diffusion,

$$\Delta t \leq \frac{h^2}{2d \max(\kappa_\Phi, \kappa_v, \kappa_S)} =: \Delta t_{\text{diff}}.$$

Overall: $\Delta t \leq \min(\Delta t_{\text{adv}}, \Delta t_{\text{diff}})$.

Proof. Von Neumann analysis on 1D advection-diffusion: amplification factor $|g| \leq 1$ requires both conditions. $\square$

Theorem 15.3 (Consistency and Convergence). *The scheme is second-order consistent in space and first-order in time. Under CFL, converges to weak solution at rate $O(h + \Delta t)$.*

Proof. Taylor expansion and Lax equivalence theorem for linear systems; extends to quasilinear via freezing coefficients. $\square$

Corollary 15.4 (Computational Falsifier). *If high-resolution simulations ($h \rightarrow 0$) show persistent grid-scale oscillations in S despite $\kappa_S > 0$, then numerical diffusion is insufficient to ensure monotonicity-preserving fluxes (e.g., WENO).*

16 Observables and Coherence Evolution

Coherence $\mathcal{C}(t)$ quantifies persistent meaning carried by rotational flow.

§17 Boundedness of Coherence Measure

Assumption 16.1. $\Phi \geq 0$, $\mathbf{v} \in L^\infty(0, T; [L^2(\Omega)]^d)$, $\nabla \times \mathbf{v} \in L^\infty(0, T; [L^2(\Omega)]^d)$.

Theorem 16.2 (Global Bound on Coherence).

$$0 \leq \mathcal{C}(t) \leq \left(\int_\Omega \Phi dx \right) \cdot \sup_t \|\nabla \times \mathbf{v}(t)\|_{L^2(\Omega)}.$$

Proof. By Hölders inequality:

$$\mathcal{C}(t) = \int \Phi \cdot \|\nabla \times \mathbf{v}\| \leq \|\Phi\|_{L^1} \cdot \|\nabla \times \mathbf{v}\|_{L^\infty}.$$

Bound $\|\nabla \times \mathbf{v}\|_{L^\infty} \leq C \|\nabla \mathbf{v}\|_{L^2}$ via Sobolev embedding in $d = 3$. $\square$

Theorem 16.3 (Time Evolution of Coherence).

$$\frac{d\mathcal{C}}{dt} \leq \int \Phi |\nabla \times (-\nabla \Phi)| + \kappa_v \int \Phi |\nabla \times \Delta \mathbf{v}| + \text{transport terms}.$$

Proof. Differentiate under integral, apply (3), integrate by parts. $\square$

Corollary 16.4 (Cognitive Falsifier). *If EEG microstate duration correlates negatively with $\mathcal{C}$ (computed from fMRI-derived $\mathbf{v}$), but not with $\int \Phi$, then rotational flow is causal for cognitive persistence.*

§18 Coherence Decay in Quiescent Regimes

Assumption 16.5. $\sigma \rightarrow 0, \nabla\Phi \rightarrow 0$ on $[T, \infty)$.

Theorem 16.6 (Exponential Coherence Decay). *Under theorem 16.5,*

$$\mathcal{C}(t) \leq \mathcal{C}(T)e^{-\nu(t-T)}, \quad \nu = \min(\kappa_v \lambda_1, \kappa_\Phi \lambda_1),$$

where λ_1 is the first eigenvalue of $-\Delta$.

Proof. Vorticity equation from curl of (3):

$$\partial_t(\nabla \times \mathbf{v}) = -(\mathbf{v} \cdot \nabla)(\nabla \times \mathbf{v}) + \kappa_v \Delta(\nabla \times \mathbf{v}).$$

Multiply by $\Phi \geq 0$, integrate, use Poincaré. □

Corollary 16.7 (Institutional Falsifier). *If cultural memes in isolated societies decay slower than predicted by ν , then external $\mathbf{v}$ (migration, media) sustains vorticity.*

17 Redshift Calibration and Cosmological Tests

§19 Calibration of α_S from Supernova Data

Assumption 17.1. Observed redshift $z_{\text{obs}} = z_{\text{ACDM}} + \delta z_{\text{RSVP}}$, with $\delta z_{\text{RSVP}} \ll 1$ at low z .

Theorem 17.2 (Linear Approximation). *For small entropy flux,*

$$z_{\text{RSVP}} \approx \alpha_S \int_{t_e}^{t_o} \frac{1}{|\Omega|} \int_{\Omega} \sigma(\Phi, \mathbf{v}) dx dt.$$

Proof. Taylor expansion of exponential in (22). □

Theorem 17.3 (Parameter Estimation). *Let $z_i \pm \sigma_i$ be N supernova redshifts. The maximum likelihood estimator is*

$$\hat{\alpha}_S = \frac{\sum_i (z_i - z_{\text{ACDM},i}) \cdot \mathcal{F}_i}{\sum_i \mathcal{F}_i^2},$$

where $\mathcal{F}_i = \int_{t_e}^{t_o} \langle \sigma \rangle dt$ is the predicted entropy flux integral.

Proof. Gaussian likelihood, linear model in α_S . □

Corollary 17.4 (Falsification Threshold). *Reject RSVP if $|\hat{\alpha}_S| > 0$ with $p < 10^{-5}$ (Bonferroni-corrected over $N \sim 1000$), or if $\hat{\alpha}_S < 0$.*

18 CLIO Architectures and Neural Implementation

§20 CLIO Loss in Differentiable Physics

Assumption 18.1. Φ is a neural field (e.g., SIREN), $\mathbf{v}$ is output of a CNN, S is computed via Monte Carlo entropy estimation.

Theorem 18.2 (Gradient Flow in Function Space). *The CLIO update is a gradient flow:*

$$\frac{d\Phi}{dt} = -\nabla_{\Phi} \mathcal{L} \in T_{\Phi} \mathcal{F},$$

where $\mathcal{F}$ is the manifold of neural fields.

Proof. Direct from (21) and functional derivative. $\square$

Theorem 18.3 (Sparsity via L^1 Proxy). *Replacing $\|\Phi\|_0$ with $\lambda_{\text{sparse}}\|\Phi\|_1$ yields*

$$\|\Phi\|_1 \leq \frac{\mathbb{E}[S] - \lambda_{\text{flow}}\|\mathbf{v}\|_2}{\lambda_{\text{sparse}}}.$$

Proof. Convex relaxation; strong duality at optimum. $\square$

Corollary 18.4 (Ablation Falsifier). *Train CLIO-net on MuJoCo. If removing $-\lambda_{\text{flow}}\|\mathbf{v}\|_2$ causes policy collapse (variance $\rightarrow 0$), agency reward is necessary.*

19 TARTAN Generalization and Out-of-Distribution Robustness

§21 Gradient-Weighted Noise Enhances Generalization

Assumption 19.1. Data manifold has curvature bounded by K , noise injected as $\Phi \leftarrow \Phi + \eta_s$, $\eta_s \sim \mathcal{N}(0, \sigma_S^2(\nabla\Phi))$.

Theorem 19.2 (Fisher Information Alignment). *Gradient-weighted noise maximizes expected Fisher information:*

$$\mathbb{E}[\|\nabla_{\theta} \log p(\Phi + \eta_s | D)\|^2] \propto \int |\nabla\Phi|^2 \cdot \sigma_S^2(\nabla\Phi) d\Phi.$$

Proof. Score function variance is Fisher; weighting by gradient concentrates sampling near high-curvature regions. $\square$

Theorem 19.3 (OOD Generalization Bound). *With gradient noise, test error satisfies*

$$Err_{OOD} \leq Err_{IID} + O\left(\frac{K}{\sqrt{n}}\right),$$

tighter than uniform noise by factor $\propto 1/|\nabla\Phi|$.

Proof Sketch. PAC-Bayesian bound with noise-induced regularization; gradient weighting reduces effective dimension. $\square$

Corollary 19.4 (Vision Transformer Falsifier). *If TARTAN noise on ViT yields worse ImageNet-C robustness than uniform noise, then data manifold is flatter than assumed.*

20 Spherepop Monte Carlo and Institutional Dynamics

§22 Monte Carlo Estimation of Merge Probability

Assumption 20.1. $\Sigma_i \sim \mathcal{U}[0, 1]$, $R_t \sim \exp(\gamma)$, $H \sim \mathcal{N}(\mu_H, \sigma_H^2)$.

Theorem 20.2 (Merge Probability).

$$P(\text{merge}) = \mathbb{P}(\Sigma_1 + \Sigma_2 > \Theta \cap R_t + 2^H < P_{\text{max}}).$$

Closed form via convolution of distributions.

Proof. Independent random variables; joint density integration. $\square$

Theorem 20.3 (Critical Permeability). *There exists Σ_c such that $P(\text{merge}) > 0.5$ iff $\Sigma > \Sigma_c(\Theta, P_{\max}, \mu_H)$.*

Proof. Solve quantile equation. □

Corollary 20.4 (Historical Falsifier). *If pre-1900 merger rate of firms exceeds $P(\text{merge})$ with $H = 0$ (no crypto), then ritual cost $\gamma(t)$ was overestimated.*

21 Philosophical Implications

§23 Ontic vs. Epistemic Interpretation of Fields

Conjecture 21.1 (Ontic Scalar). *Φ is a physical field (e.g., quantum information density), conserved like charge.*

Conjecture 21.2 (Epistemic Vector). *v is inferential flow in an active inference agent, not spacetime velocity.*

Theorem 21.3 (Duality). *The RSVP dynamics are invariant under observer reinterpretation: $(\Phi, v, S) \mapsto (p(\text{belief}), \nabla \log p, -\log p)$.*

Proof. Active inference free energy is $\mathcal{L}$; gradient flow matches CLIO. □

Corollary 21.4 (Philosophical Falsifier). *If Φ violates Bell inequality but S does not, then Φ is ontic, S epistemic.*

22 Historical Lineage

§24 From Dissipative Structures to Semantic Physics

Theorem 22.1 (Lineage Isomorphism). *Prigogines $\dot{S}_i = \sigma \geq 0 \leftrightarrow \text{RSVP (4) with } \lambda = 0$.*

Theorem 22.2 (FristonRSVP Duality). *CLIO loss $\mathcal{L} = \mathbb{E}[S] + \text{complexity} \leftrightarrow \text{variational free energy}$.*

Proof. $S = -\log p$, sparsity = model complexity. □

Corollary 22.3 (Unification Falsifier). *If a system violates both second law and free energy principle, RSVP substrate is falsified.*

23 Future Directions and Open Problems

§25 Open Conjectures and Experimental Roadmap

Conjecture 23.1 (Nonlocal Extension). *Replace Δ with $(-\Delta)^s$, $s \in (0, 1)$, to model long-range semantic coupling.*

Conjecture 23.2 (Quantum RSVP). *Promote $\Phi \rightarrow \hat{\Phi}$, $v \rightarrow \hat{v}$, $S \rightarrow \hat{\rho} \log \hat{\rho}$ on Hilbert space.*

Theorem 23.3 (Experimental Priority). *The following triplet is jointly testable:*

$$(z_{\text{RSVP}}, \Gamma_{\text{fMRI}}, P_{\text{merge}})$$

via cosmology, neuroscience, and institutional data.

Proof. All derived from same $(\Phi, \mathbf{v}, S)$ via coarse-graining functors F_i . $\square$

Corollary 23.4 (Final Falsifier). *If any two observables decorrelate after controlling for the third, the unified ontology is broken.*

A Categorical Coarse-Graining

Define categories RSVP, EBSSC, etc., with functors preserving entropy monotonicity and sparsity.

B Numerical Implementation

Pseudocode with CFL condition:

$$\Delta t \leq \min \left(\frac{h^2}{2d \max(\kappa_\Phi, \kappa_v, \kappa_S)}, \frac{h}{\|\mathbf{v}\|_\infty} \right).$$

C Linear Stability Analysis

Linearize around uniform state $(\Phi_*, \mathbf{v}_* = 0, S_*)$. Dispersion relation:

$$\lambda = -(\kappa_\Phi + \kappa_v)k^2 \pm ik\Phi_*.$$

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